

SHARATH B S, M.Sc., Ph.D.

Computational Chemist · AI Drug Discovery · Cheminformatics

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PROFESSIONAL SUMMARY

Computational Chemist with 4+ years of postdoctoral and industry experience, specializing in cheminformatics, molecular modeling, and AI-assisted drug discovery. Strong analytical and problem-solving skills applied to biological data analysis, scientific computing, and workflow development and management. Proven scientific productivity with experience evaluating research literature and producing high-quality scientific writing. Actively exploring agentic AI and retrieval-augmented generation (RAG) to translate complex biological datasets into actionable insights.

PROFESSIONAL EXPERIENCE

Research Scientist

Dec 2025 – Present

UR Advanced Therapeutics (BioVaram) | Hyderabad, India

- Build computational pipelines for rational peptide design, improving structural-stability and function prediction to guide candidate selection.
- Deploy ML models for peptide property prediction, reducing sequence-design and screening cycle time.
- Design and optimize self-assembling and cell-penetrating peptides (CPPs) with advanced MD simulations to support biomaterial development.
- Lead multi-omics analysis (lipidomics, proteomics, transcriptomics) for exosome characterization and biomarker prioritization.

Postdoctoral Researcher

Feb 2024 – Mar 2025

Pharmacoinformatics Research Group, Universität Wien | Vienna, Austria

Project: Roche ROADS — In silico predictive & translational safety

- Mined large-scale preclinical datasets in CDISC SEND format to support regulatory-ready translational safety analyses.
- Developed and deployed knowledge-based ML models to predict clinical outcomes and safety profiles from preclinical toxicity data.
- Partnered with toxicologists to map off-target effects and toxicity endpoints, enabling data-driven decisions in drug development.

Postdoctoral Researcher

Aug 2021 – Dec 2023

Soongsil University | Seoul, South Korea

Project: In silico quorum-sensing inhibitor screening using cheminformatics tools (NRF, South Korea)

- Engineered end-to-end cheminformatics pipelines for high-throughput virtual screening, accelerating lead discovery from large compound libraries.
- Automated data extraction, processing, and proteochemometric modeling with reproducible Python workflows.
- Applied ligand- and structure-based CADD to identify and prioritize high-confidence drug candidates.

Assistant Lecturer

Oct 2020 – Jun 2021

Kuvempu University | *Shivamogga, Karnataka, India* — Taught biotechnology courses and mentored students in early research career development.

Scientific Manager

Nov 2014 – Apr 2015

DiponEd BioIntelligence | *Bengaluru, India* — Managed R&D on biological formulations (PRP, PRFM) and stem-cell isolation, culture, and maintenance.

EDUCATION

Ph.D. in Biotechnology

Feb 2019

Kuvempu University | Karnataka, India

Thesis: Drug target identification through dynamics of protein interaction network analysis using VEGF-induced monocytes

- Pathway modelling, network pharmacology, and power-graph analysis; molecular docking, MD simulation, and binding free-energy calculation; experimental validation of bottleneck proteins with small-molecule compounds.

M.Sc. in Biotechnology

Jun 2014

Sahyadri Science College, Kuvempu University | Karnataka, India

Project: Phytosynthesis, characterization, and induced toxic effect of zinc nanoparticles on mitotic chromosomes of Allium cepa L.

- Nanoparticle characterization (UV-Vis, FTIR, AFM, HR-TEM) and cytotoxicity assessment on Allium cepa L. root tips.

B.Sc. in Chemistry, Biotechnology & Zoology

Jun 2013

Sahyadri Science College, Kuvempu University | Karnataka, India

CERTIFICATIONS

- Machine Learning and Molecular AI — Uresearcher LLC (Nov 2021 – Jul 2022). ML/DL for drug discovery with hands-on capstone.
- Data Analysis for Genomics — HarvardX / edX (Apr 2023 – Jul 2023). Bioconductor, functional genomics case studies, and advanced Bioconductor with capstone.
- Version Control of a Python Project using Git — Coursera (Mar 2023).

TECHNICAL EXPERTISE

- **Computational & Molecular Modeling:** MD simulations (GROMACS, Amber); coarse-grained MD (martini); enhanced sampling (steered MD, umbrella sampling, replica exchange); membrane-protein MD; free-energy calculations (MM/GBSA, MM/PBSA); molecular docking; protein/peptide modeling; interaction analysis (PyMOL, ChimeraX, VMD).
- **Cheminformatics & Molecular AI:** Molecular descriptors/fingerprints (ECFP, MACCS, RDKit); pharmacophore & QSAR modeling.
- **Data Science & MLOps:** Scikit-learn, TensorFlow, PyTorch; feature engineering; dimensionality reduction (PCA, t-SNE, UMAP); model interpretability (SHAP). Strong interest in agentic AI and RAG for developing data-to-insight workflows in bioinformatics.
- **Systems Biology & Network Analysis:** Multi-omics analysis (lipidomics, proteomics, transcriptomics); pathway analysis (KEGG, Reactome); PPI networks (Cytoscape, STRING); gene ontology enrichment; Bioconductor.
- **Omics Data Analysis Tools:** Bioconductor & its packages - SRA Toolkit, FastQC, FastQ Screen, MultiQC, Cutadapt, Trim Galore, STAR, Bowtie2, kallisto, RSEM, Samtools, Subread, R, DESeq2.
- **Programming & HPC:** Python, Bash, Git/GitHub; large-scale simulations on HPC clusters (SLURM, PBS); SSH/SCP for remote workflow management.

SELECTED PUBLICATIONS

Total: 27 publications · Citations: 312 · h-index: 10 · Full list: [Google Scholar](#)

1. Sharath B. S., Kandagalla S., Lee J.* Development of machine learning models based on molecular fingerprints for selectivity of novel small molecules against JAK2 protein. J. Comput. Chem. 2023 (IF 4.8). <https://doi.org/10.1002/jcc.27103>
2. Sharath B. S., Rimac H., Lee J.* In-silico screening of quorum sensing inhibitor candidates obtained by chemical similarity search. Molecules 2022 (IF 4.6). <https://doi.org/10.3390/molecules27154887>
3. Sharath B. S., Kandagalla S., Vikas H. M., Pavan Kumar G. S., Prabhakar B. T.*, Manjunatha H.* A systems biology approach to identify the key targets of curcumin and capsaicin that downregulate pro-inflammatory pathways in human monocytes. Comput. Biol. Chem. 2019 (IF 3.1). <https://doi.org/10.1016/j.compbiolchem.2019.107162>

4. Kandagalla S.†, Sharath B. S.†, Rimac H.* , Grishina M., Manjunatha H.* , Potemkin V.* Computational insights into the binding mode of curcumin analogues against EP300 HAT domain as potent acetyltransferase inhibitors. J. Mol. Graph. Model. 2020 (†Equal contribution; IF 3.0). <https://doi.org/10.1016/j.jmgm.2020.107756>
5. Sharath B. S., Kandagalla S., Manjunatha H.* A network pharmacology approach to investigate the pharmacological effect of curcumin and capsaicin targets in cancer angiogenesis by module-based PPI network analysis. J. Proteins Proteomics 2019. <https://doi.org/10.1007/s42485-019-00012-y>

SELECTED CONFERENCES & PRESENTATIONS

- International Conference on Biological Physics (ICBP 2023), Seoul, South Korea (Aug 14–18, 2023).
Presented title: “Computational Approach to Predict Off-target Binding Profile for Virtual Small Molecule Screening in Rational Drug Designing.”
- Lecture Notes in Computer Science (LNCS): Advances in Bioinformatics and Computational Biology (BSB 2020).
Presented title: “A systems biology driven approach to map the EP300 interactors using comprehensive protein interaction network.”

WORKSHOPS

- 2023 Annual Conference of the Korean Society for Bioinformatics (BIOINFO 2023), Yeosu, South Korea (Nov 13–15, 2023).
- Hands-on Training in Genomics and Transcriptomics — jointly organized by The University of Trans-Disciplinary Health Sciences and Technology (TDU) and Bangalore Genomics Centre (BGC), Centre for Functional Genomics and Bioinformatics, Bengaluru, India.
- International Workshop on Computational Techniques for Wetlab Data Analysis (IWCWA), National College (Autonomous), Tiruchirappalli, India (Sep 22–24, 2015).

REFERENCES

Available upon request.